

ADP Z-axis Manual

——ADP Series

Product Model:

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1 Product Overview

The Z-axis is used to carry the movement of SADP products in the vertical direction.

1.1 Z-axis Main Features

- ◆ Support vertical movement with minimum center spacing of 9mm.
- ◆ Movement speed can be up to 180 mm/s
- ◆ Provide a vertical thrust of not less than 3kg
- ◆ Loaded with a 410g weight, it is capable of achieving vertical self-locking in the event of a power failure.
- ◆ Movement accuracy could be within $\pm 0.2\text{mm}$.
- ◆ Support 232 and CAN control, at the same time in coordination with SADP, it could support 485 control
- ◆ Support single-channel, 2-channel, 4-channel and 8-channel application with SADP16
- ◆ Support single-channel, 2-channel and 4-channel application with SADP18
- ◆ Support single-channel application with SADP20 or SADP28

2 Product Performance

2.1 Performance Index

Maximum speed: 180mm/s

Maximum stroke: 180mm

Movement maximum noise: $\leq 65\text{dBA}$

Movement accuracy: $\pm 0.2\text{mm}$

Maximum thrust: 3kg

Full stroke lifespan: 1 million times

2.2 Z180 Series Product Appearance

2.2.1 3D drawing

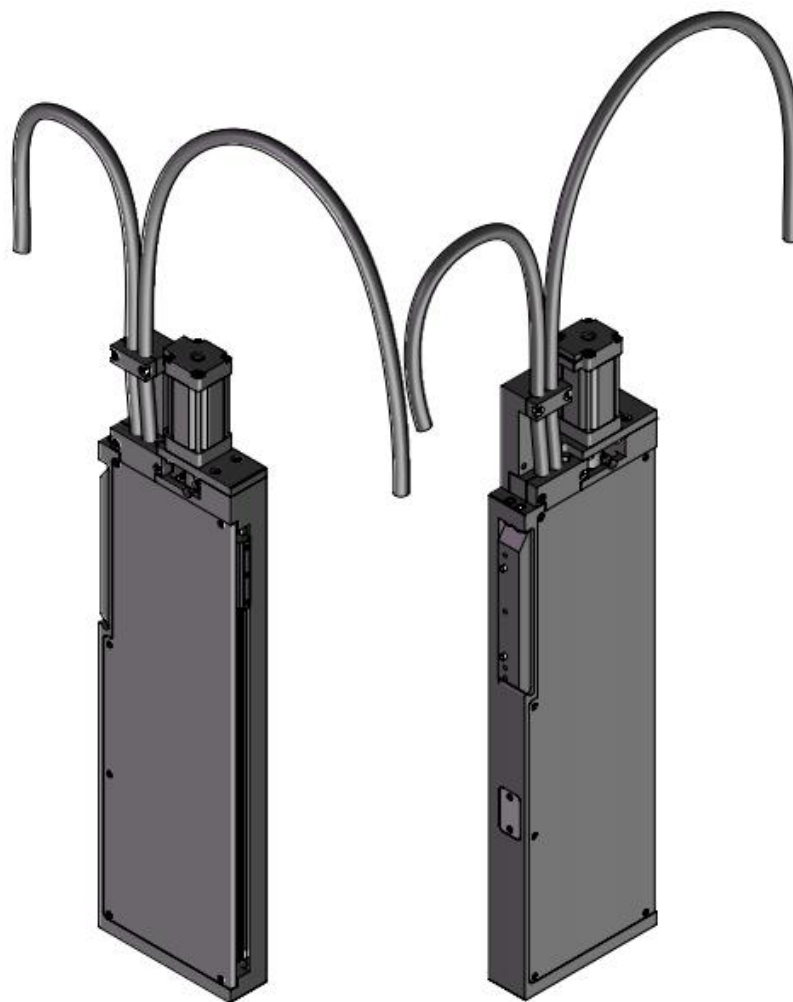


Figure 1 Z180 3D drawing

2.3.2 Overall dimension

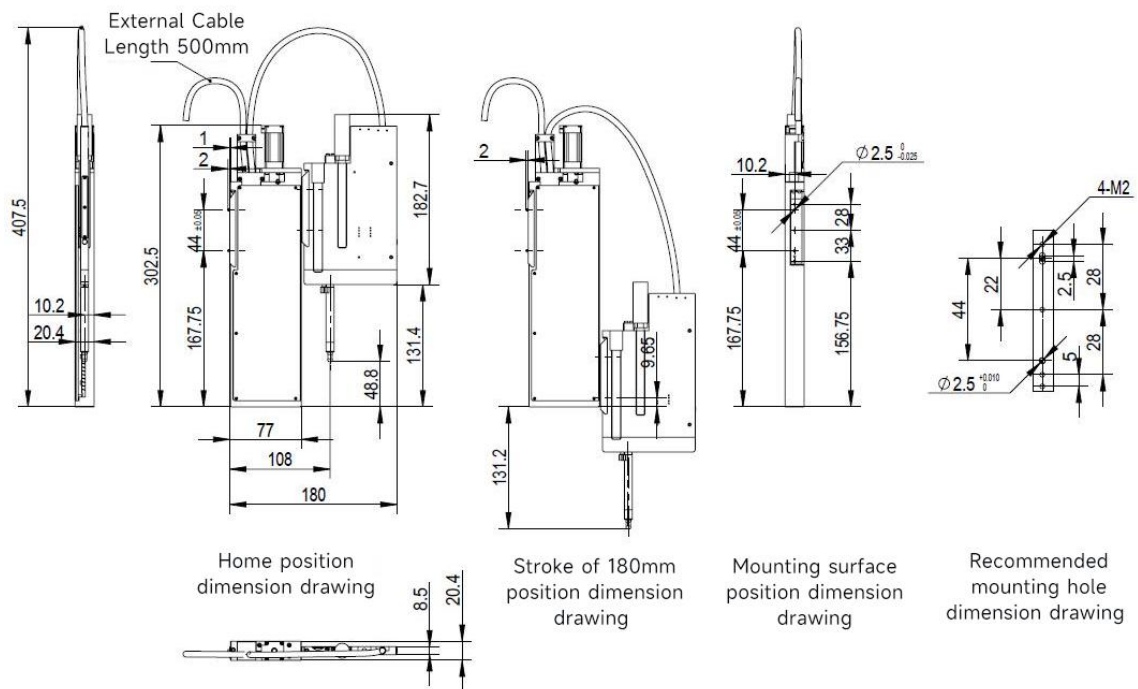


Figure 4 SADP16+Z180 (Single) Overall dimension

3 Product Electrical Interface

3.1 Z-axis Interface Lead Definition

The Z-axis is generally integrated with the SADP and shares 232, 485 and CAN communication. When using the device, communication lines that are not needed should undergo insulation treatment to prevent functional abnormalities caused by the core contact.

Using highly flexible drag chain cable to connect to the external control system with AWG26 cores. The following table defines the communication cable:

Function	Cable color	Remarks
DC 24V+	Red	Power input $\pm 1\%$, $\geq 1\text{A}$
GND	Black	Grounding
RS232-RX	Green	Communication Interface
RS232-TX	Blue	Communication Interface
RS485A	Orange	Communication Interface
RS485B	White	Communication Interface
CAN L	Yellow	Communication Interface
CAN H	Yellow-green	Communication Interface

4 Communication Protocol

4.1 Communication Method

4.1.1 Communication interface

The Z-axis is controlled with the same interface as the SADP supporting the following methods:

- ◆ RS232
- ◆ RS485
- ◆ CAN

4.1.2 Protocol

Z-axis supports both serial and CAN communication. the serial port supports DT and OEM Protocol, with 8 data bits, no parity, and 1 stop bit. Both protocols can be used to control command flow, enabling compatibility with SADP. The specific protocol format is the same as that of SADP.

- ◆ DT Protocol
- ◆ OEM Protocol
- ◆ CAN Protocol

Notice: When using serial communication, the suggested transmission interval is 10ms, following a request-response communication mode.

DT Protocol

This protocol is based on RS232 and RS485 communication, the communication protocol has no check bit, you can easily use the serial interface debugging tool to send a string to control the Z-axis. Data is sent and received as ASCII string, which is convenient for users to debug. The communication protocol has no check bit so there is a risk of data loss, please refer to the DT protocol for details.

OEM Protocol

This protocol is based on RS232 and RS485 communication, the communication protocol includes communication sequences and check byte, which can effectively prevent data loss. During the working period, the controller polls the Z-axis status and analyses the queried status to decide whether the Z-axis completed commands execution or there is an error, please refer to the OEM protocol for details.

CAN Protocol

This communication protocol is used for local area network CAN communication. It enables the control of the Z-axis and the setting of parameters by reading and writing registers without checking the Z-axis status. The Z-axis will automatically upload the current status whenever there is a change, please refer to the CAN protocol for details.

DT Protocol Format

Whenever the Z-axis receives a string, it parses the string and verifies whether the address matches and the syntax is correct. It then returns the status of the first command execution. In the case of multiple commands, querying the status is necessary to confirm the successful execution of subsequent commands. The status can also be queried for commands with empty data in the sending data area.

Data Format

Table 1 DT command protocol format for transmission

Sequence number	Function	Number of bytes	Description
1-2	Address	2	The target communication address should be set as a unique address number for each drive on the bus, with a value range of 1-32. Normal communication occurs only when the address matches; otherwise, received commands will be ignored
3	Direction	1	The character > indicates sending data
4+n	Data	n	ASCII commands strings, see operation commands for details.
5+n	Terminator	1	The carriage return character 0x0D, every communication data should end with a carriage return.

Table 2 DT return data protocol format

Sequence number	Function	Number of bytes	Description
1-2	Address	2	The original device address, which matches the address sent, indicates which address returned the data.
3	Direction	1	The character < indicates return data
	Status	2	See status table
	Special symbols:	1	' ' if no return data this symbol is not displayed
4+n	Data	n	Return data string, see the corresponding return data of the operation commands for details
5+n	Terminator	1	The carriage return character 0x0D, every communication data should end with a carriage return

4.2 OEM Protocol Format

The communication protocol includes communication sequences and check byte, which can effectively prevent data loss, so it is recommended to use OEM protocol to communicate with Z-axis. During operation, the controller queries the Z-axis status by sending a blank command, analyzes the queried status to determine whether the Z-axis executes a completed command or an error occurs, and the commands to send the data in the data area is empty can query the status.

Commands format for transmission

Table 3 OEM command protocol format for transmission

Data fields	Data type	Number of bytes	Description
Frame header	UInt8	1	Fixed value 0XAA, indicating the start of data
Address	UInt8	1	The communication address, each drive on each bus should be set to a unique address number, taking values from 1 to 32. It will communicate properly when the addresses match, otherwise it will ignore the received commands.
Data length	UInt8	1	Data area data length
Data	Char	n	ASCII commands strings, see operation commands for details
Checksum	UInt8	1	The 8-bit checksum is calculated from the frame header to the end byte of the data area. The value obtained takes the last 8 bits of data.

Return data format

Table 4 OEM return data protocol format

Data fields	Data type	Number of bytes	Description
Frame header	UInt8	1	Fixed value 0X55, indicating the start of data
Address	UInt8	1	Communication address, each drive on each bus should be set to a unique address number, the address is the original address of the device when returning data
Status	UInt8	1	Current module status, see status table
Data length	UInt8	1	The data length of the data area, when the data area is empty, the data length is 0
Data	Char	n	ASCII code return data, see operation commands for details
Checksum	UInt8	1	The 8-bit checksum is calculated from the header of the frame to the end byte of the data area. The value obtained takes the last 8 bits of data

4.3 CAN Protocol Format

Sending data format

Frame type: data frame

Message identifier type: extended frame

ID: use extended ID, please refer to the following table

DLC: data length 8

Data field: the data length is the fixed 8 bytes, including communication Sequence number, register address, and data as shown in the table below.

ID data format

bit23-bit16	bit15-bit8	bit7-bit0
Commands	Original address	Target address

Data area data format

Table 5 CAN Protocol commands format

byte7	byte6-byte5	byte4	byte3-byte0
Communication sequence number	Object index	Object sub-index	4 bytes of data

Commands: Used to indicate the type of operation to change the frame data, as shown in the following table:

Commands	Function	Description
0x00	Responding	Read and write data return
0x01	Write	Write object dictionary Return value: status, see status table
0x02	Read	Read object dictionary, if there is no corresponding object dictionary, no data is returned.
0x03	Process data	Used for uploading real time data that does not require an answer, e.g. status change active uploading send should via this commands
0x04	Heartbeat	The heartbeat data uploaded at regular intervals can be used to detect whether the device is online or not, the heartbeat uploaded data is the node status, the status information is shown in the status table
0x80	Alarm	The device error is actively reported in the commands, the error information is shown in the status table

Original address: The sender address.

Target address: The receiver address.

Communication sequence number: It is used to distinguish which frame of data is sent and received. It is recommended that users add 1 to the communication sequence number before each frame of data is sent so that each frame of data is different.

Object index: The object index is 16-bit index address data, which is used to define the object dictionary table, the function of the object dictionary table is shown in the object dictionary table, and the details are shown in the CAN Object Dictionary Chapter.

Object sub-index: The sub-index is an 8-bit sub-index address data, which is used together with the object index to define the object dictionary table.

Data: A data length of 32 bits represents the communication data, which is a signed integer. For detailed information please refer to the CAN Object Dictionary chapter.

5 Serial Interface Commands

This chapter describes the data format in the data area of the OEM and DT protocols, which represents the format of operational instructions. This data is in ASCII string format and allows the simultaneous transmission of multiple sets of instructions. The Z-axis and SADP will parse and execute the command sets one by one. These commands are classified by function as follows:

- ◆ Initialization commands
- ◆ Control commands
- ◆ Parameter read/write commands
- ◆ Query commands
- ◆ System control commands

5.1 Commands Syntax

<CMD><n1,n2,n3><CMD><n1,n2,n3><CMD><n1,n2,n3><CMD><n1,n2,n3>

The ASCII format of the data returned is as follows.

<n1,n2,n3>

Among them:

<CMD> indicates commands, represented by letters from a to z and A to Z, with a maximum of two letters. See command details

<n1,n2,n3> indicates the command parameters. Multiple parameters are separated by commas ','. Commands with no parameters can be empty. If specific parameters need to be set to default values, the corresponding parameter before the comma can be left blank. If the remaining parameters are all blank, they can be omitted. For example, the command Zp1000 with the last parameter blank represents that the last parameter is empty.

Notice:

1)<> is used to distinguish data blocks and does not need to be transmit.

2)Commands are case-sensitive

3)The command consists of a maximum of 2 letters, which starts with an uppercase letter followed by a lowercase letter. A single-letter command is represented in uppercase. Special characters '?' represent a query command, '{}' represents a loop control command. A standalone uppercase letter command is for system control, and the command of an uppercase letter followed by a lowercase letter is for operation control.。

5.2 Status

Each instruction comes with a return status indicating whether the instruction was successfully executed, if the device encountered errors, and its current state (busy, idle, etc.). The status is represented by a single-byte hexadecimal data. Only the Z-axis-related statuses are described here; for SADP statuses, refer to the SADP manual, as they share a common status:

We agreed:

0-9: Working

10-19: Command execution error

20-49: Alarm, aspiration and dispense of liquid is allowed after the alarm.

>=50: Fault, need to troubleshoot and reinitialize

Table 6 Status

Value	Function	Description
0	Idle	Device idle status
1	Busy	The device is under operation
2	Successfully executed	Status of "successfully executed" the command
...		
10	Parameter overrun	Commands parameters out of range
11	Parameter error	Commands parameter error
12	Syntax error	Commands syntax error

13	Invalid command	Commands do not support
14	Address error	Read and write register address error
15	Write prohibit	This address is not allowed to be written to
16	Read prohibit	This address is not allowed to be read out
18	Z-axis uninitialized	Z-axis is not initialized
19	Z-axis unconnected	Z-axis is not connected
20	Fail to pick up TIP	The TIP is not picked up even after the maximum stroke
...		
80	Motor stall error	Need to reinitialize and troubleshoot
81	Z-axis motor drive failure	Need to reinitialize and troubleshoot
82	Z-axis motor optocoupler error	Need to reinitialize and troubleshoot
83	Z-axis storage error	Need to reinitialize and troubleshoot

5.3 Command Details

This section describes the detailed individual command, and the mounting functions are divided into the following categories.

- ◆ Initialization commands
- ◆ Control commands
- ◆ Parameter setting commands
- ◆ Query commands
- ◆ Circulation control commands

Notice:

[] indicates optional parameters, if the optional parameters are empty it will be executed according to the default parameter in the protocol.

The content inside <> is a command, the mark <> does not need to be sent, it is just used to distinguish letters and commands.

5.3.1 Z-axis control commands

5.3.1.1 <Zz>n1 Z-axis initialization

Power-up and initialize the Z-axis to find the top 0 position. Other movements are only allowed after the initialization is completed. It is recommended to use the default values for the initialization speed.

Table 7 Zz command

Command value	Parameters	Data range	Unit	Default value	Description
Zz	n1	0~18000 0	um/s	50000	Initialization speed

Return data: see Status Table

5.3.1.2 <Zp>n1,[n2] Move to specified position

Control the Z-axis move to the specified position.

Table 8 Zp command

Command value	Parameters	Data range	Unit	Default value	Description
Zp	n1	0~18000 0	um	0	Position
	[n2]	0~18000 0	um/s	50000	Speed

Return data: see Status Table

5.3.1.3 <Zu>n1,[n2] Move upward by a relative distance

Control the Z-axis move upward to the specified position relative to the current position.

Table 9 Zu command

Command value	Parameters	Data range	Unit	Default value	Description
Zu	n1	0~18000 0	um	0	Relative distance position
	[n2]	0~18000 0	um/s	50000	Operating speed

Return data: see Status Table

5.3.1.4 <Zd>n1,[n2] Move downward by a relative distance

Control the Z-axis move upward to the specified position relative to the current position.

Table 10 Zd command

Command value	Parameters	Data range	Unit	Default value	Description
Zd	n1	0~18000 0	um	0	Relative distance position
	[n2]	0~18000 0	um/s	50000	Operating speed

Return data: see Status Table

5.3.1.5 <Zg>n1,[n2] Move downward to pick up TIP

Control the Z-axis to move downward, and when the TIP is detected, the motor stops due to a stall. The duration of Zg movement should be greater than 0.3s; otherwise, premature triggering may result in a failure to pick up the TIP. It is recommended to use the default speed for picking up the TIP (at default speed, the ADP probe head is more than 1cm away from the TIP port). The power should be set between 80-100%. At this setting, the force for picking up the head is approximately 30N.

Table 11 Zg command

Command value	Parameters	Data range	Unit	Default value	Description
Zg	n1	0~180000	um	50000	Operating speed
	[n2]	0~100		80	Power percentage

Return data: see Status Table

5.3.1.6 <Zt>Stop Z-axis

Immediately stop Z-axis movement.

5.3.2 Parameter Setting**5.3.2.1 Register**

The registers are used for user configuration and viewing of ADP parameters, facilitating flexible application by users.

Table 12 Parameter register

Address	R/W	Data range	Unit	Default value	Description
81	R/W	100/125/250/500/1000		500	CAN baud rate, requires a reboot to take effect.
82	R/W	0-1		0	Active reporting after motion completion 0: Do not report 1: Report
89	R/W	0-1		0	Address switch, when set to 0, the default address is 1
90	R/W	0-255		1	The default address is 1. After connecting to ADP, it becomes ADP address +40. Connecting

					to ADP prohibits modifying the address. To change the address, write 1 to register 89.
94	R/W	9600\19200\38400\115200		38400	Serial Port Baud Rate
100	R			0	Z-axis current status (see status table)
101	R		um		Z-axis current position
102	R/W		mA	900	Set Z-axis rated current
103	R/W	0-1		1	Set Z-axis liquid level emergency stop 0: GPIO stop 1: GPIO+ communication stopped
104	R		mm		Set the maximum stroke of Z-axis
105	R		mm/s	180	Set the maximum speed of Z-axis
106	R		um	6000	Z-axis lead
107	R/W		ms	1000	Heartbeat interval
108	R/W		ustep/s ²	160000	Acceleration
109	R/W		ustep/s ²	160000	Deceleration
110	R/W	0-1		0	Motor stall detection 0: Turn off motor blockage detection. 1: Turn on motor blockage detection.
113	R/W		ustep		Initial calibration compensation value
114	R/W	0-1		0	Holding Force Switch (Effective when 131 is written as 1) 0: Enable holding force 1: Disable holding force
119	R/W	0-1		1	CAN Communication Mode Switch 1: Peer Mode 0: Master-Slave Mode (Liquid Handling Master)
120	R/W	0-255		1	Device Address
121	R				Software version
122	R				Device model, same for each SADP-Z
123	R				Device sequence number
124	R				Hardware version number
130	R/W	0-32		3	Holding current
131	R/W	0-1		0	Self-locking mode 0: Mechanical self-locking mode 1: Programmatic self-locking mode
132	R/W		ustep	0	Manual compensation calibration value: Used to adjust the consistency of the Z-axis in multi-axis systems. Height can be fine-tuned based on the value of register 113. 553ustep is 1mm. Example: Lower the Z-axis by 1mm. 1. Read the value of register 113, assuming it is 347.; 2. Write the value 1106 to register 132 3. Initialize; 4. Calibrate. (Ensure there is enough space for the load to move through the entire range during calibration, and be mindful of collision risks)

5.3.2.2 <Wr>n1,n2 Write register

Write register.

Table 13 Wr command

Command value	Parameters	Data range	Unit	Default value	Description
Wr	n1	81~131	N/A	N/A	Register address
	n2				Write data

5.3.2.3 <Rr>n1,[n2] Read register

Read registers, read the specified number of registers from the starting address, read registers larger than 1, return data separated by ','.

Table 14 Rr command

Command value	Parameters	Data range	Unit	Default value	Description
Rr	n1	81~131	N/A	N/A	Start address
	n2				Number of registers

Return data: the status part is shown in the status table, the data area part is the data to be fetched, and the return data of multiple addresses are separated by ','.

5.3.3 System operation commands

5.3.3.1 Query Status

<?>

Query status via command ?

Return data: see status table

5.3.3.2 Loop control command

The loop control commands are used to control the cycle of the command string, the loop can be nested, a maximum of 20 cycles including nested cycles are supported in a command string.

{ Loop start position

Comm and value	Parameters	Data range	Unit	Default value	Description
{	N/A		N/A	0	Start loop

}[n1]Loop end position

Comm and value	Parameters	Data range	Unit	Default value	Description
}	[n1]	0-2147483647		0	No parameter or 0: infinite loop Other values: loop times

{:loop start

}:loop ends, followed by the loop count. If there is no loop count or the data is empty, it indicates an infinite loop.

Note: loop can be nested

5.3.3.3 <L>n1 Delay

The delay is used for internal system delay, mainly as a delay between two command executions.

Table 15 L command

Comm and value	Parameters	Data range	Unit	Default value	Description
L	n1	0~2147483647	ms	N/A	Delay time

Return data: see Status Table

5.3.3.4 <U> Rest command

This command is used to reset the device.

Table 17 U command

Comm and value	Parameters	Data range	Unit	Default value	Description
U	N1			123456	The parameter must be 123456 for the successful reset

5.3.3.5 <M>n1 Restore factory settings command

After executing the commands, the device needs to be restarted to take effect.

Table 18 M command

Comm and value	Parameters	Data range	Unit	Default value	Description
M	N1			123456	The parameter must be 123456 for the successful restoration to the factory settings.

5.3.3.6 <S> Power down to save setting parameters

After executing the commands, the modified register parameters will be saved when power down.

6 CAN Object Dictionary

The object dictionary includes 8-bit main index and 8-bit sub-index, different operations and parameter read/write are indicated by ordered index numbers, with all data having both read and write permissions. The control commands include initialization, aspirate, dispense, etc., all follow different main indexes to indicate different commands, and sub-indexes are different parameters, the number of parameters is the same as the number of serial command parameters, for example, the serial interface commands It[n1][n2][n3], for instance, need to modify the default parameters before executing the commands.

6.1 Z-axis Control Commands

Function	Main index	Sub-index	R/W permissions	Data range	Default value	Description
Zero setting	0x4100	0	W	1-18000 0 (um/s)	Required	Initialization, speed during parameter initialization process
Move to absolute position	0x4101	0	W	0-18000 0(um)	Required	Counting from the top to the bottom after initialization
		1	RW	0-18000 0 (um/s)	Required	Move to absolute position speed
Move upward to relative position	0x4102	0	W	0~18000 0 (um)	Required	Move up from current position
		1	RW	0-18000 0 (um/s)	50,000	Upward motion speed
Move downward to relative position	0x4103	0	W	0~18000 0 (um)	Required	Move down from current position
		1	RW	0-18000 0 (um/s)	50,000	Downward motion speed
Move downward to pick up TIP	0x4104	0	W	0~18000 0 (um/s)	Required	Downward motion speed
		1	RW	0-100	80	Power percentage
Emergency stop	0x4108	0	W	0	0	Stop movement
calibration	0x9000	0	W	0	0	zero-calibration
Power down save mode	0x9F10	0	W	0	0	Power down to save data
Restore factory settings	0x9F10	1	W	123456	0	Restore factory settings
Active reporting	0x9F00	5	RW	0-1	0	Active reporting of movement completion
Reset	0x9F00	3	W	123456	0	Reboot

6.2 W/R Register

We have agreed upon the main index 0x2000 for internal registers within ADP, and the sub-index

represents the register address. The register addresses can be found in Table 11, which details parameter registers.

6.3 Alarm Data

The Z-axis sends alarm information through command 0x80, and the format of the alarm data is the same as the error information in the status query.

7 Application Process

When using serial communication, due to DT protocol is lacking of data verification, it is better to use it for debugging. For devices, it is recommended to use the OEM protocol. The data format of the data area in the OEM protocol is the same as the DT protocol, only the frame header and frame tail data formats are different. When applying the OEM protocol, it is essential to query the status after each operation to ensure that the command execution is completed before proceeding to the next step. Taking the address 41 as an example:

7.1 DT Protocol Application Example

7.1.1 Initialization

	Data	Description
Tx	41>Zz50000	41: Target device address; 50,000: initialize the running speed to 50,000 $\mu\text{m/s}$
Rx	41<2	41: Target device address; 2: successful command execution status(see status table for others)

7.1.2 Move to specified position

	Data	Description
Tx	41>Zp50000,180000	41: Target device address; 50,000: move to the absolute position of 50,000 μm ; 180,000: running speed is 180,000 $\mu\text{m/s}$.
Rx	41<2	41: Target device address; 2: successful command execution status(see status table for others)

7.1.3 Move downward to pick up TIP

	Data	Description
Tx	41>Zg50000,80	41: Target device address; 50,000: move to the absolute position of 50,000 μm ; 80: pick up TIP with 80% power; used in coordination with ADP for picking up the TIP.
Rx	41<2	41: Target device address; 2: successful command execution status(see status table for others)

7.1.4 Query current status

	Data	Description
Tx	41>?	41: Target device address; ?: used to query the current device status, equal to query status register 100
Rx	41<0	41: Target device address; 0: idle status(see status table for others)

7.1.5 Read parameter

	Data	Description
Tx	41>Rr101	41: Target device address; 101: check the Z-axis current position (Other register address, see set parameter)
Rx	41<2:0	41: Target device address; 2: successful command execution status(see status table for others); 0: current position is 0.

7.1.6 Set parameter

	Data	Description
Tx	41>Wr82,1	41: Target device address; 82: active report register ; 1: turn on the active report function, namely, actively report data after the motion is completed to determine if the motion is completed

Rx	41<2	41: Target device address; 2: successful command execution status(see status table for others)
-----------	------	--

7.2 OEM protocol application example (transmit in HEX mode)

Note: The OEM protocol is an encapsulation of the DT protocol. Please refer to the DT protocol for explanations of command strings and return strings.

7.2.1 Initialization

	Data	Description
Tx	AA 29 07 5A 7A 35 30 30 30 30 A3	AA: Frame header; 29: Target device address; 07: Data length; 5A 7A 35 30 30 30 30: Command string of 'Zz50000'; A3: Frame Checksum
Rx	55 29 02 00 80	55: Frame header; 29: Target device address; 02: successful command execution status(see status table for others); 00: Return string length; 80: Frame Checksum

7.2.2 Move to specified position

	Data	Description
Tx	AA 29 0E 5A 70 35 30 30 30 30 2C 31 38 30 30 30 30 F5	AA: Frame header; 29: Target device address; 0E: Data length; 5A 70 35 30 30 30 30 2C 31 38 30 30 30: Command string of 'Zp50000,180000'; F5: Frame Checksum
Rx	55 29 02 00 80	55: Frame header; 29: Target device address; 02: successful command execution status(see status table for others); 00: Return string length; 80: Frame Checksum

7.2.3 Move downward to pick up TIP

	Data	Description
Tx	AA 29 0A 5A 67 35 30 30 30 30 2C 38 30 27	AA: Frame header; 29: Target device address; 0A: Data length; 5A 67 35 30 30 30 30 2C 38 30: Command string of 'Zg,50000,80'; 27: Frame Checksum
Rx	55 29 02 00 80	55: Frame header; 29: Target device address; 02: successful command execution status(see status table for others); 00: Return string length; 80: Frame Checksum

7.2.4 Query current status

	Data	Description
Tx	AA 29 01 3F 13	AA: Frame header; 29: Target device address; 01: Data length; 3F: Command string of '?'; 13: Frame Checksum
Rx	55 29 00 00 7E	55: Frame header; 29: Target device address; 00: idle status(see status table for others); 00: Return string length; 7E: Frame Checksum

7.2.5 Read parameter

	Data	Description
Tx	AA 29 05 52 72 31 30 31 2E	AA: Frame header; 29: Target device address; 05: Data length; 52 72 31 30 31: Command string of 'Rr101'; 2E: Frame Checksum
Rx	55 29 02 01 30 B1	55: Frame header; 29: Target device address; 02: successful command execution status(see status table for others); 01: Return string length; 30: Data area(current position is 0); B1: Frame Checksum

7.2.6 Set parameter

	Data	Description
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Tx	AA 29 06 57 72 38 32 2C 31 69	AA: Frame header; 29: Target device address; 06: Data length; 57 72 38 32 2C 31: Command string of 'Wr82,1'; 69: Frame Checksum
Rx	55 29 02 00 80	55: Frame header; 29: Target device address; 02: successful command execution status(see status table for others); 00: Return string length; 80: Frame Checksum

7.3 CAN Protocol Application Example

Note: please refer to Section 6.1 for control commands. Each control command has a unique main index and multiple sub-indices. Send frame data with a non-zero sub-index first, and finally, send frame data with a sub-index of zero. SADP starts moving upon receiving a command with a sub-index of zero.

7.3.1 Initialization

	ID	Data	Description
Tx	00 01 00 29	01 41 00 00 00 00 c3 50	00 01: Write command; 00: Source device address; 29: Target device address; 01: Communication sequence number; 41 00: Main index (initialization control); 00: Sub-index (initialization speed); 00 00 c3 50: Data (parameter setting of initialization speed is 50000 um/s).
Rx	00 00 29 00	01 41 00 00 00 00 00 02	00 00: Response data; 29: Responding device address; 00: Receiving response data device address; 01: Communication sequence (consistent with the received command sequence); 41 00: Main index; 00: Sub-index; 00 00 00 02: Status (successful command execution, see status table for others)

7.3.2 Move to specified position

	ID	Data	Description
First Tx	00 01 00 29	01 41 01 01 00 02 bf 20	00 01: Write command; 00: Source device address; 29: Target device address; 01: Communication sequence number; 41 01: Main index (move to specified position); 01: Sub-index (position parameter: move to specified position); 00 02 bf 20: Data (speed parameter data of moving to specified position is 50000 um/s).
Rx	00 00 29 00	01 41 00 00 00 00 00 02	00 00: Response data; 29: Responding device address; 00: Receiving response data device address; 01: Communication sequence (consistent with the received command sequence); 41 00: Main index; 00: Sub-index; 00 00 00 02: Status (successful command execution, see status table for others)
Next Tx	00 01 00 29	01 41 01 00 00 00 c3 50	00 01: Write command; 00: Source device address; 29: Target device address; 01: Communication sequence number; 41 01: Main index (move to specified position); 01: Sub-index (position parameter: move to specified position); 00 00 c3 50: Data (speed data of moving to specified position is 50000 um/s).
Rx	00 00 29 00	01 41 00 00 00 00 00 02	00 00: Response data; 29: Responding device address; 00: Receiving response data device address; 01: Communication sequence (consistent with the received command sequence); 41 00: Main index; 00: Sub-index; 00 00 00 02: Status (successful command execution, see status table for others)

7.3.3 Move downward to pick up TIP

	ID	Data	Description
First Tx	00 01 00 29	01 41 04 01 00 00 00 50	00 01: Write command; 00: Source device address; 29: Target device address; 01: Communication sequence number; 41 04: Main index (move downward to pick up TIP); 01: Sub-index (position parameter: move to specified position); 00 00 00 50: Data (move downward to pick up TIP with 80% power).
Rx	00 00 29 00	01 41 04 01 00 00 00 02	00 00: Response data; 29: Responding device address; 00: Receiving response data device address; 01: Communication sequence (consistent with the received command sequence); 41 00: Main index; 00: Sub-index; 00 00 00 02: Status (successful command execution, see status table for others)
Next Tx	00 01 00 29	01 41 04 00 00 00 c3 50	00 01: Write command; 00: Source device address; 29: Target device address; 01: Communication sequence number; 41 04: Main index (move downward to pick up TIP); 00: Sub-index (speed parameter of moving downward to pick up TIP); 00 00 c3 50: Data (speed parameter data of moving downward to pick up TIP is 50,000 μ m/s).
Rx	00 00 29 00	01 41 04 00 00 00 00 02	00 00: Response data; 29: Responding device address; 00: Receiving response data device address; 01: Communication sequence (consistent with the received command sequence); 41 00: Main index; 00: Sub-index; 00 00 00 02: Status (successful command execution, see status table for others)

7.3.4 Read parameter

	ID	Data	Description
Tx	00 02 00 29	01 20 00 65 00 00 00 00	00 02: Read command; 00: Source device address; 29: Target device address; 01: Communication sequence number; 20 00: Main index (read register main index); 65: Sub-index (current position register, register 101); 00 00 00 00: Data (read parameter data area).
Rx	00 00 29 00	01 20 00 65 00 00 00 00	00 00: Response data; 29: Responding device address; 00: Receiving response data device address; 01: Communication sequence (consistent with the received command sequence); 20 00: Main index; 65: Sub-index; 00 00 00 00: Register parameter value (Current position is 0).

7.3.5 Set parameter

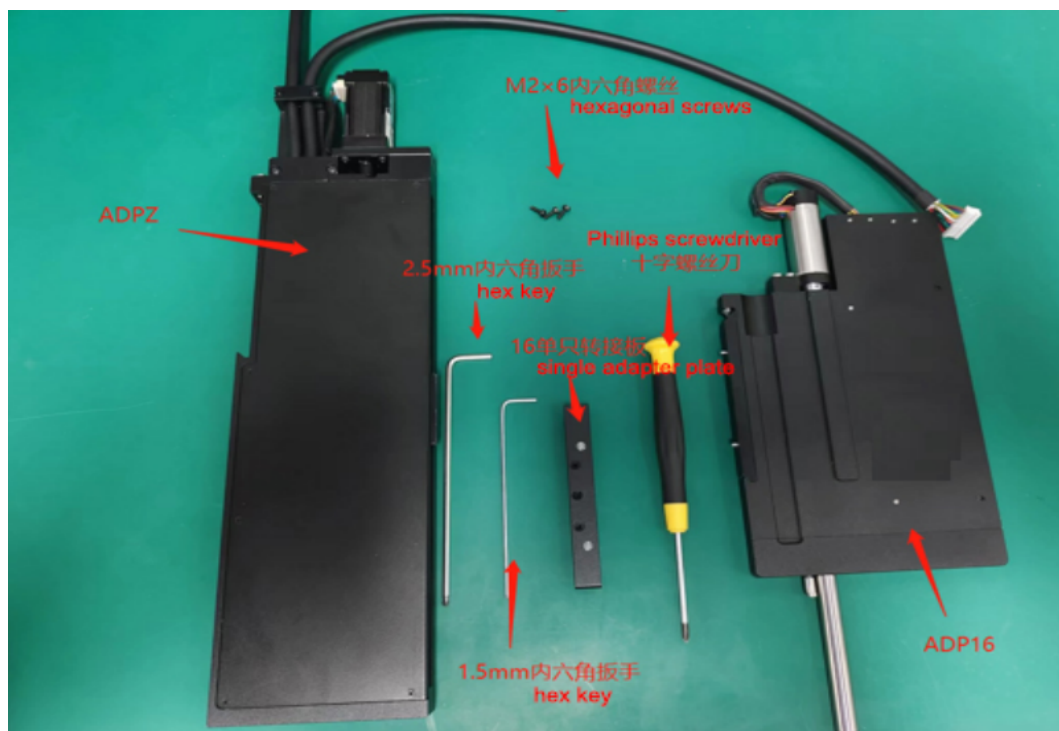
	ID	Data	Description
Tx	00 01 00 29	01 20 00 52 00 00 00 01	00 01: Write command; 00: Source device address; 29: Target device address; 01: Communication sequence number; 20 00: Main index (read register main index); 52: Sub-index (actively report register, register is 82); 00 00 00 01: Data (set the value of register 82 to 1, turn on the active report function).

Rx	00 00 29 00	01 20 00 52 00 00 00 02	00 01: Response data; 29: Responding device address; 00: Receiving response data device address; 01: Communication sequence (consistent with the received command sequence); 20 00: Main index; 52: Sub-index; 00 00 00 02: Register parameter value(setting successfully, see status table for others)
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8 ADPZ and ADP Installation Process

8.1 Installation process of ADPZ and ADP16

Step 1: Prepare the following tools and equipment.



Step 2: Install the ADP16 single adapter plate with the blunt side facing down on the ADPZ.





Step 3: Remove the wire pressboard from ADP16, insert connection cable between ADP and ADPZ into ADP, then install the wire pressboard (please note that the pressboard should be level with the surface of ADP)



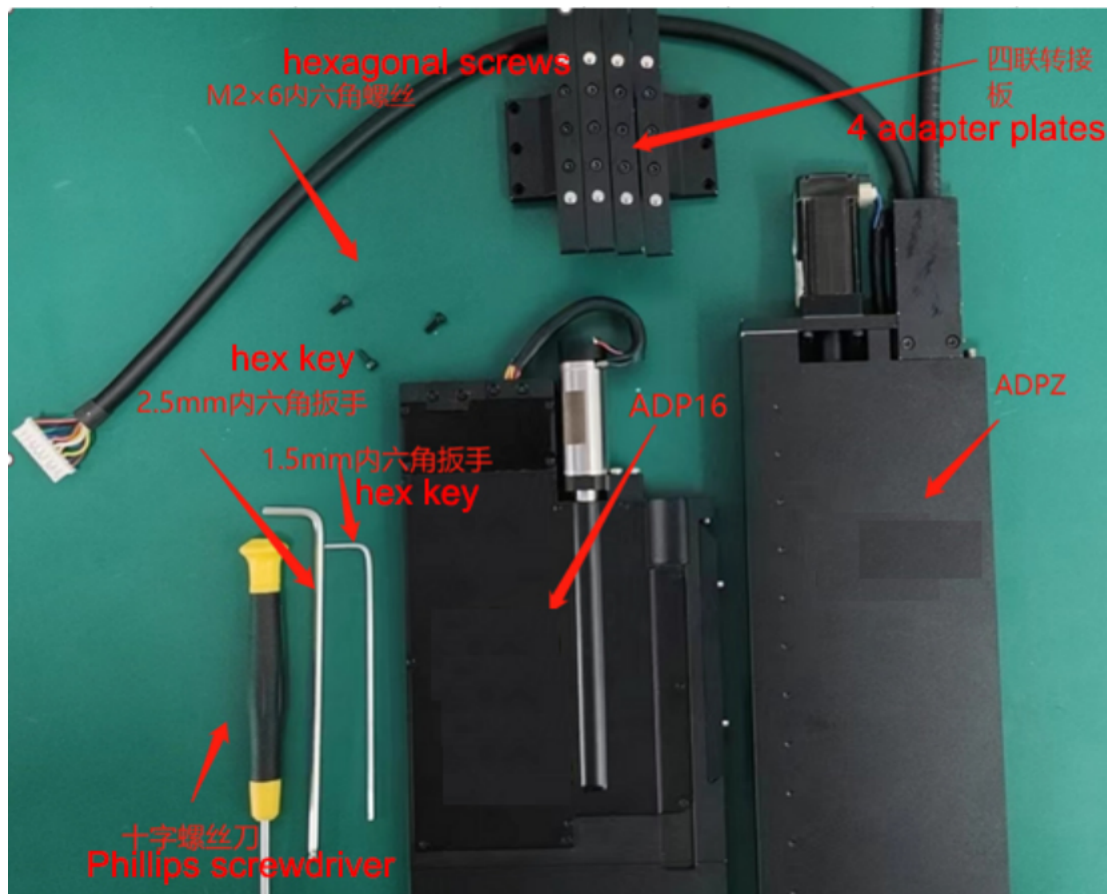


Step 4: Remove the mounting plate of ADP16, and then fix ADP16 on 16 single adapter plates (the pin of ADP16 single adapter plate should be aligned with the hole of ADP16 during installation)



8.2 Installation Process of ADPZ and ADP16 Fixed in 4-channel Movement Mechanism

Step 1: Prepare the following tools and equipment



Step 2: Arrange 4 adapter plates in order and install them on ADPZ

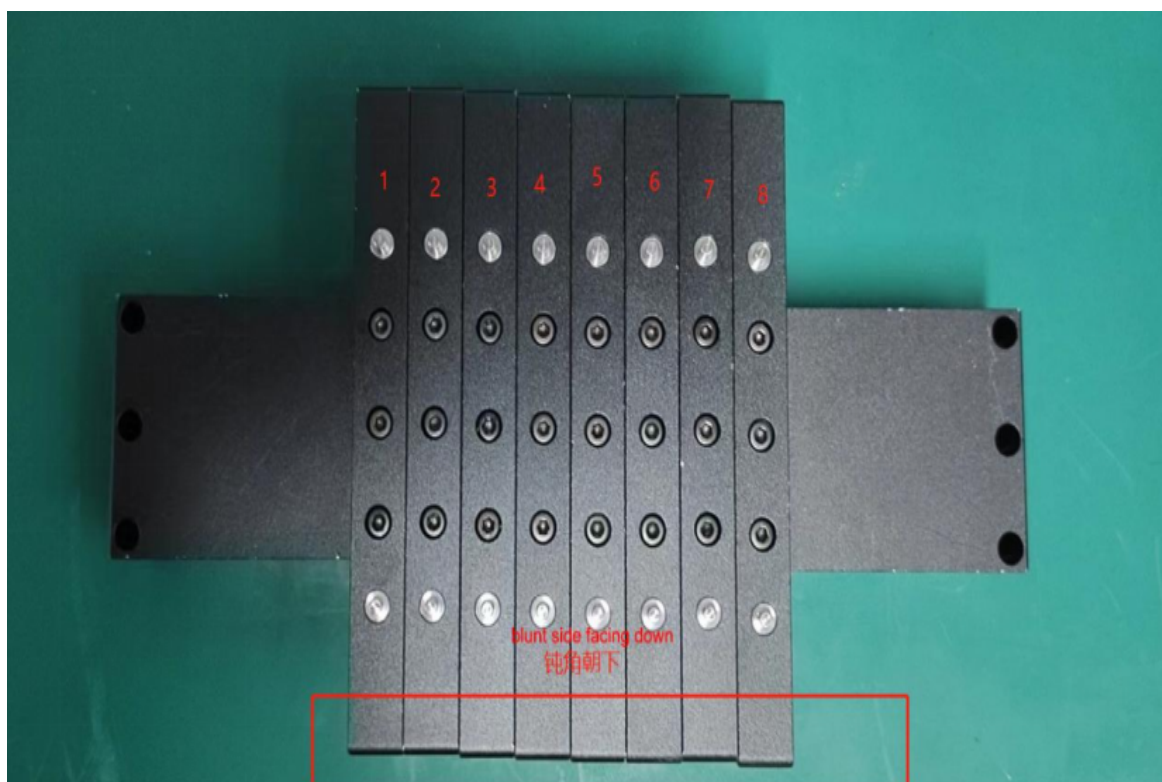


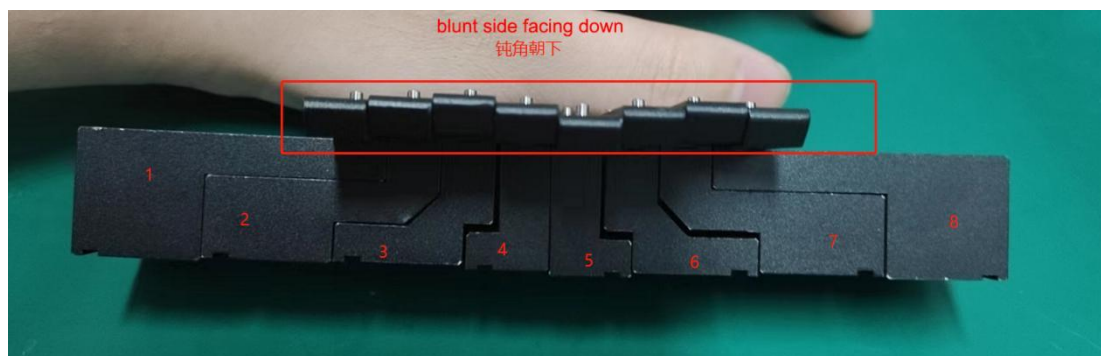


For other installation steps, please refer to the installation process of ADPZ and single ADP16.

8.3 Installation Process of ADPZ and ADP16 Fixed in 8-channel Movement Mechanism

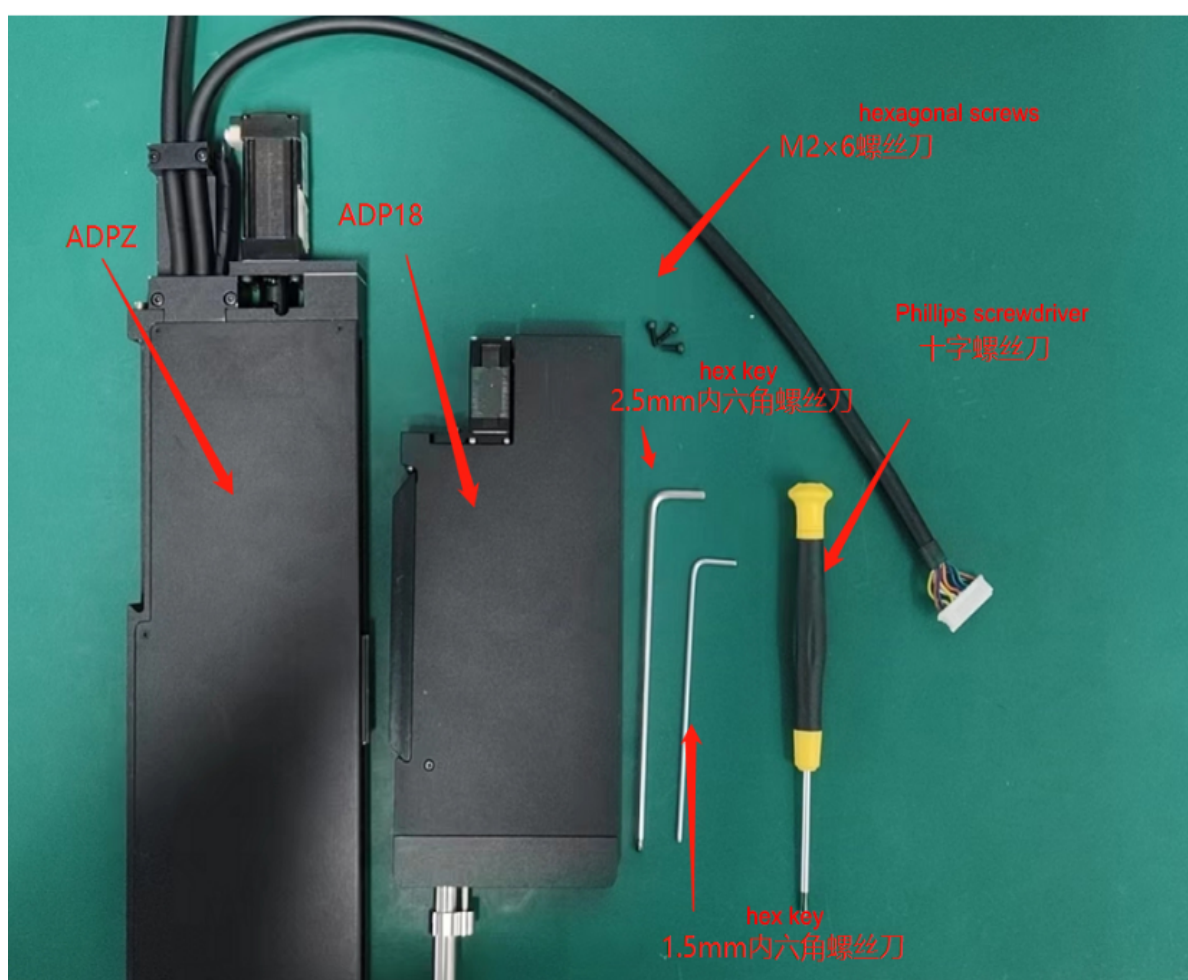
The difference between the 8-channel and 4-channel installation is that there are 4 more adapter plates. For other installation steps, refer to the 4-channel installation process



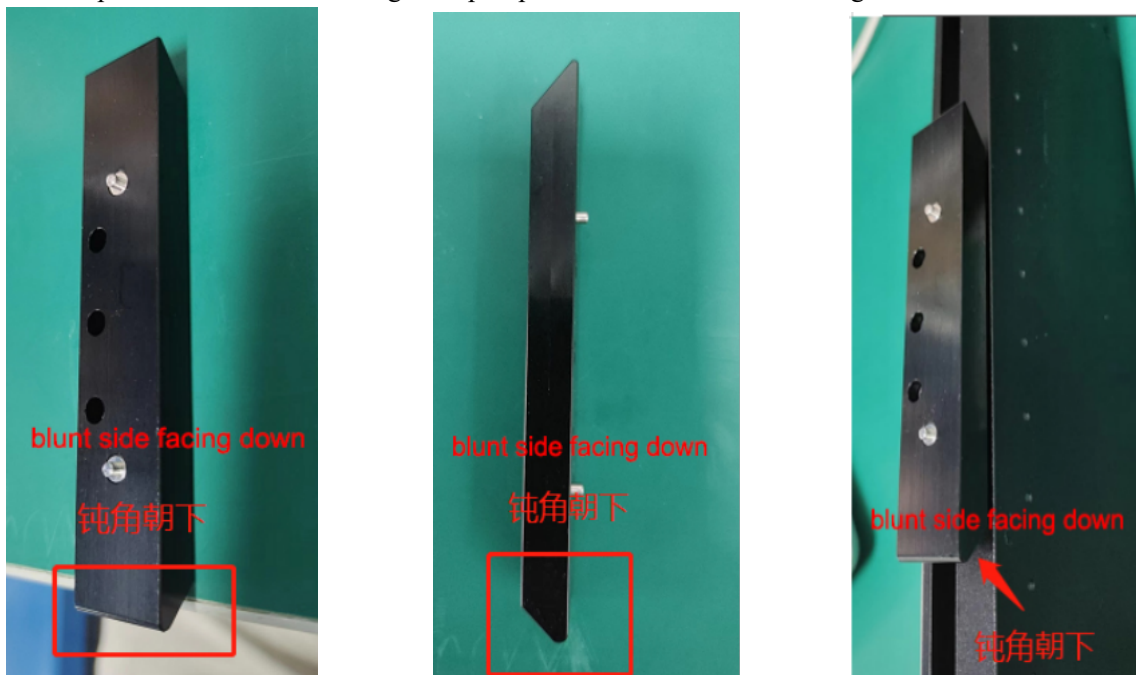


8.4 Installation Process of ADPZ and ADP18

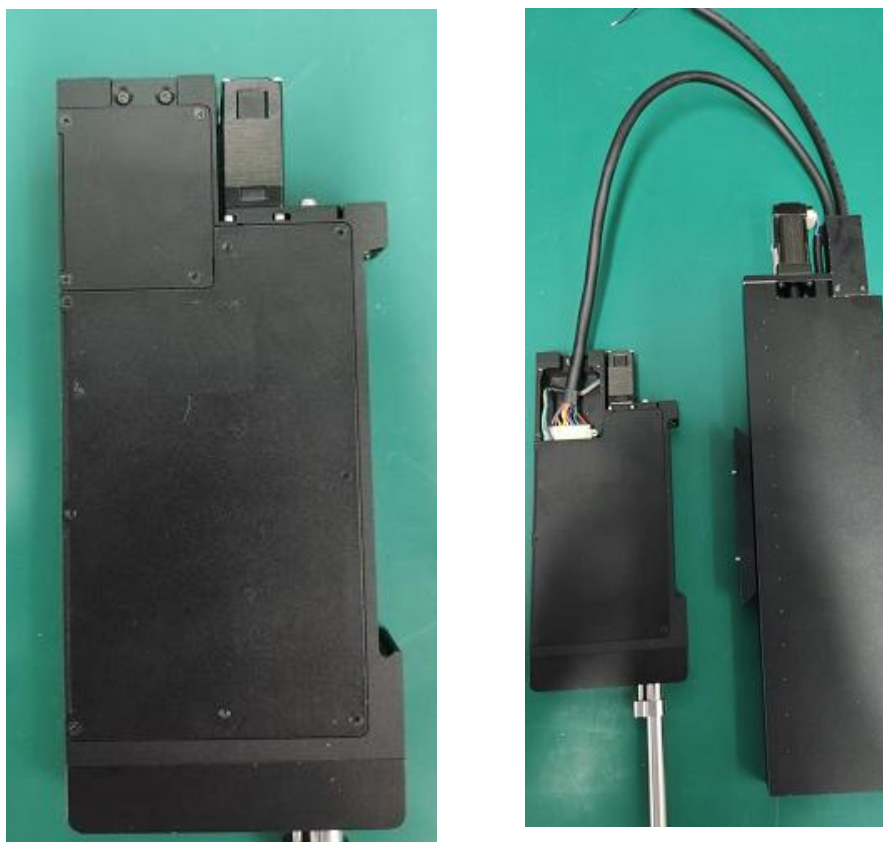
Step 1: Prepare the following tools and equipment.



Step 2: Install the ADP18 single adapter plate with the blunt side facing down on the ADPZ.



Step 3: Remove the wire pressboard from ADP18, insert connection cable between ADP and ADPZ into ADP, then install the wire pressboard (please note that the pressboard should be level with the surface of ADP)



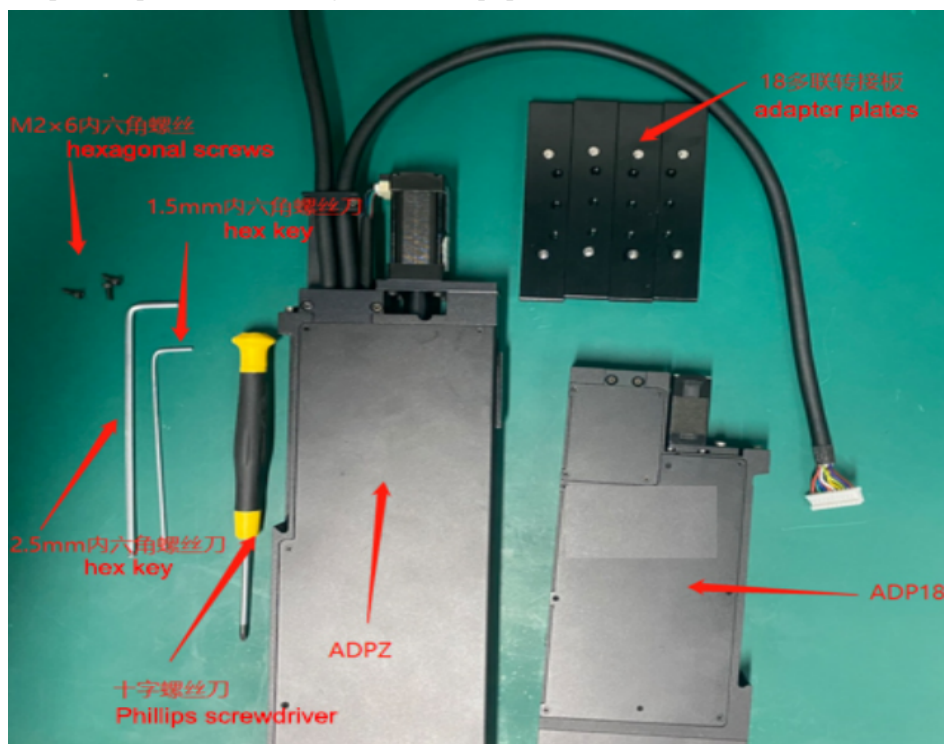


Step 4: Remove the mounting plate of ADP18, and then fix ADP18 on 18 single adapter plates (the pin of ADP18 single adapter plate should be aligned with the hole of ADP18 during installation).



8.5 Installation Process of ADPZ and ADP18 Fixed in 4-channel Movement Mechanism

Step 1: Prepare the following tools and equipment.



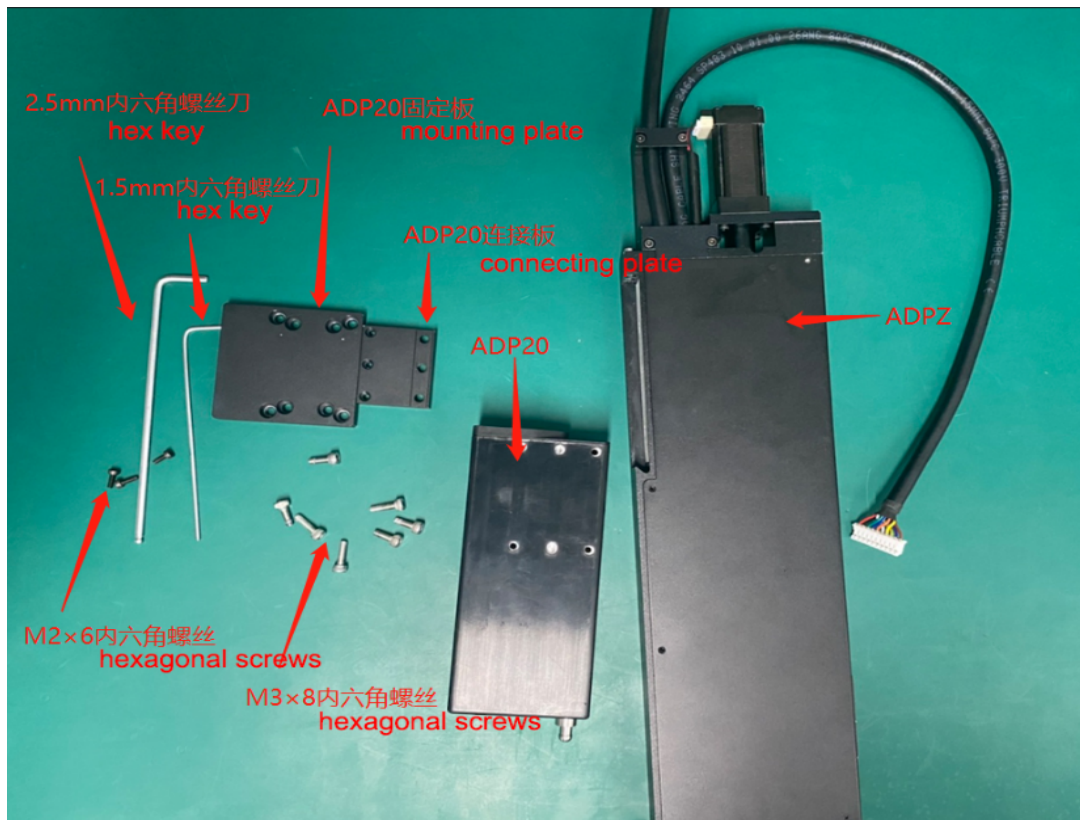
Step 2: Arrange 4 adapter plates in order and install them on ADPZ.



For other installation steps, please refer to the installation process of single ADP16.

8.6 Installation Process of ADPZ and ADP20

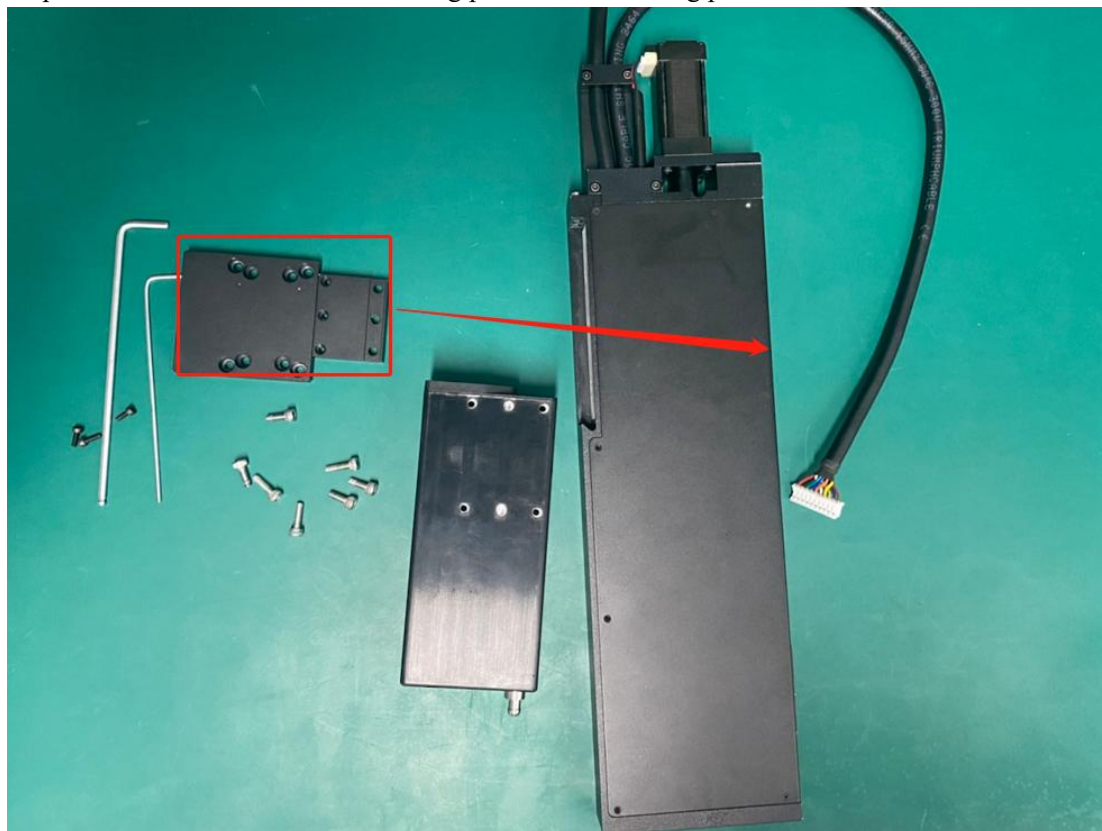
Step 1: Prepare the following tools and equipment.



Step 2: Connect the ADP20 connecting plate and mounting plate



Step 3: Connect the connected mounting plate and connecting plate to ADPZ



Step 4: Fix ADP18 on the connecting plate, and connect in the ADPZ and ADP cables



9 Application Suggestion

When using the 8-channel ADP, it must be ensured that the independent lifting stroke is not greater than 2.5cm after folding; otherwise, mechanical limitations may lead to equipment damage.

10 Common Faults and Troubleshooting Methods

Note: The following descriptions are troubleshooting methods that arise when performed in a compliant operating environment and under rated operating conditions.

	Failure	Reason	Troubleshooting
1	Input commands do not work	The drive has reported an error	Re-powering
2	Location is inaccurate or input does not match	Input error	Refer to the specification and re-enter the correct commands
		Signal interference	Check if the serial interface cables are in good condition
		Loose motor cable contact	Plugging the motor cable house tightly
		Rotation is prevented by external resistance	Examine the external interference force
		Not upgraded to the latest version of the program	Upgrade procedure (please consult your engineer if you need to upgrade)
3	Initialize the Z-axis to move downward, or an 82 error occurs upon power-up	Loose motor cable contact	Check the motor wires
		Optocoupler has a problem	Check register 202, see if the optocoupler triggering is normal

11 Environmental Conditions

Item	Unit	Value
Working environment temperature	Degrees Celsius	+5°C~+40°C
Humidity of working environment	RH%	30%~80% non-condensing
Storage temperature	Degrees Celsius	-20°C~+60°C
Storage humidity	RH%	10%~85%

12 Safety Precautions

For your and other users' safety, please read the safety precautions carefully.

This manual uses the following marks. Please fully understand what they mean before reading on.

CAUTION 		Any content with this mark is a part of the user must pay attention to, otherwise, it will cause damage to the product or other losses due to improper operation.	
	Must operate as warned, with specific warnings or caution messages described within the triangle.		Actions that must be prohibited, with specific prohibitions described in circles.
	Important commands or actions must be performed.		

WARNING 		Any content with this mark, related to the safe use of the product and the user's safety, must strictly follow the requirements of the operation, otherwise, it may cause damage to the product or endanger the user's safety.	
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 CAUTION			
	Please turn off the power when it is idle for a long time or when the whole machine is repaired, otherwise, it will cause fire or electric shock.		Do not put it in wet, dusty, greasy environment or close to heat generating equipment, otherwise, it will cause product failure, even malfunction, fire or electric shock.
	Prohibit hot-swapping any serial interface cable, motor cable, optocoupler cable or valve power cable, otherwise, it will cause communication or other parts to fail.		If there is a long-term non-use of the hole please use the matching plug, otherwise, may cause impurities and airflow into the valve body and affect normal use.
	It is forbidden to disassemble the valve or adjust any parameters by yourself, otherwise, the valve may not work properly.		Prohibit dry grinding operation without liquid infiltration.

 WARNING			
	Do not disassemble Do not disassemble, repair or modify the product by yourself, otherwise, it may cause fire or		Avoid use in wet environments Moisture may cause electric shock.

	electric shock.		
	Cut off the power when abnormal If there is an abnormal situation, immediately cut off the power. Otherwise, it may cause fire or electric shock.		Protection when using corrosive fluids Strictly follow the applicability medium of the specification book to use, when using corrosive fluids must pay attention to protection.

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